

Cellular function given parametric variation in the Hodgkin and Huxley model of excitability

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How is reliable physiological function maintained in cells despite considerable variability in the values of key parameters of multiple interacting processes that govern that function? Here, we use the classic Hodgkin–Huxley formulation of the squid giant axon action potential to propose a possible approach to this problem. Although the full Hodgkin–Huxley model is very sensitive to fluctuations that independently occur in its many parameters, the outcome is in fact determined by simple combinations of these parameters along two physiological dimensions: structural and kinetic (denoted *S* and *K*, respectively). Structural parameters describe the properties of the cell, including its capacitance and the densities of its ion channels. Kinetic parameters are those that describe the opening and closing of the voltage-dependent conductances. The impacts of parametric fluctuations on the dynamics of the system—seemingly complex in the high-dimensional representation of the Hodgkin–Huxley model—are tractable when examined within the *S*–*K* plane. We demonstrate that slow inactivation, a ubiquitous activity-dependent feature of ionic channels, is a powerful local homeostatic control mechanism that stabilizes excitability amid changes in structural and kinetic parameters.

excitability | homeostasis | inactivation | dimensionality reduction

The canonical Hodgkin–Huxley mathematical model of membrane excitability is embedded in a high-dimensional parameter space. In their original report, Hodgkin and Huxley (1) indicate that the parameters vary substantially between different cells, an observation that is extensively documented by electrophysiologists who have studied excitable membranes over half a century (2, 3). This cell-to-cell intrinsic variability—that is, variability that cannot be attributed to measurement uncertainties (2, 4)—is habitually averaged out in attempts to generalize findings (5, 6).

There are many reasons to assume that cell-to-cell variability is also expressed in a given cell over time (7–13). For instance: kinetic parameters of channel gating change due to continuous modulation and activity-dependent roaming in protein configuration space; the ratio between the number of different channel proteins in the membrane might change due to differential protein expression or turnover; the membrane capacitance and leak conductance change during massive cell growth, movement, or contact of the cell with biological matrices that impact on membrane surface tension.

Randomly and independently pulled from the physiological range indicated by Hodgkin and Huxley, many combinations of parameters give rise to an excitable “solution” (i.e., stimulus-driven excitable membranes that generate single action potentials), but many other combinations lead to either nonexcitable or oscillatory, pacemaking membranes (14). The solutions are sensitive to relatively slight parametric independent variations. This is in contrast to the biological neurons and muscles that maintain relatively invariant patterns of activity that are seemingly more robust compared with the classic Hodgkin–Huxley-type models used to describe them. Cell-to-cell and within-cell parametric variation challenge our understanding of establish-

ment and maintenance of excitability, as well as the methods we use to extract parameters from voltage-clamp data and construct suitable models (2, 15–17). This essential problem goes beyond the regulation of excitability; it belongs to a class of open questions that concern the study of organization in biological systems and the emergence of macroscopic functional order from a large space of potential microscopic “disordered” configurations (18).

Attempts to account for invariant excitability given parametric variation focus on activity-dependent, homeostatic coordination of channel protein expression (11, 19–21). The interpretation is corroborated by correlations between mRNA concentrations of different ionic channel proteins (22). It is also supported by elegant simulations showing how centralized activity-dependent (feedback) regulation that controls protein expression may navigate a cell into one of many functional solutions (11, 23). However, channel protein densities are not the only determinants of membrane excitability status. Even in the relatively simple Hodgkin–Huxley model—a single compartment with two voltage-dependent conductances—more than 10 parameters are involved, possibly varying and impacting each other in a wide range of time scales (subsecond to hours and days).

We acknowledge the difference between the high-dimensional parameter space dictated by the explicit Hodgkin–Huxley model and the dimensionality of the physiological space within which regulation of excitability is embedded. In other words, we ask how many physiologically relevant dimensions are needed to capture the dynamics of excitability and its regulation in the

Significance

Macroscopic cellular function is maintained despite extensive variations in underlying elementary constituents, including the size of the cell, and the number, distribution, and kinetics of their proteins. Here, we take advantage of the sound theoretical and experimental basis of action potential generation to analyze macroscopic cellular invariance given microscopic variation. This analysis points to a significant gap between the high-dimensional level of description captured by biophysical measurements of channel function and the lower, physiological dimensionality, to which cellular function is sensitive. When examined in a lower dimension, a simple rule that relies on sodium channel slow inactivation provides a powerful homeostatic control mechanism that maintains excitability amid changes in protein concentrations and their kinetics.

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Data deposition: The Mathematica notebook related to this study has been deposited at <https://www.wolframcloud.com/objects/marom/SM.nb>.

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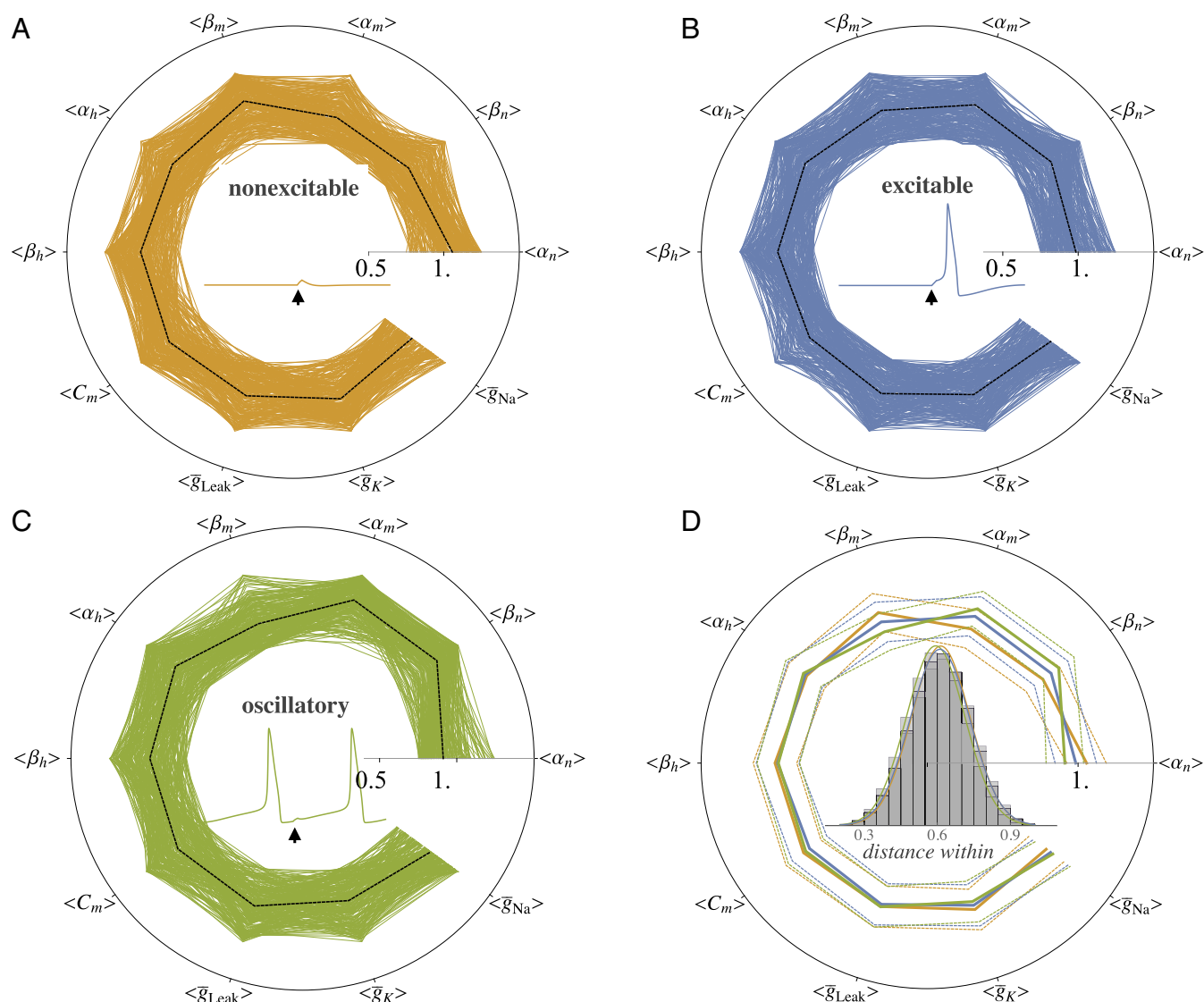


Fig. 2. Minor change in parameters significantly affects membrane excitability. Realizations (10,000) of a full Hodgkin–Huxley model; each realization is uniquely defined by a vector of 10 parameters, expressed in terms of their scaling relative to the values chosen by Hodgkin and Huxley. Responses are classified to three excitability statuses (different colors): excitable (2,225), nonexcitable (4,884), and oscillatory (2,891). Subsets of the results (200 for each excitability status) are presented in polar plots (A–C): Given a list of 10 scaling parameters, the value of each parameter is depicted along its own (angular) axis, and the entire vector is depicted as a line that connects the 10 scaling parameters. The standard Hodgkin–Huxley model would be a line passing through 1 for all scaling parameters. Mean vectors are depicted by dashed lines. The histograms in *D* depict Euclidean distance between vectors of scaled parameters within each of the three excitability classes.

Given a vector of 10 scaling parameters, the value of each parameter is depicted along its own (angular) axis, and the entire vector is depicted as one line that connects the 10 scaling parameters. The standard Hodgkin–Huxley model would be a line passing through 1 for all scaling parameters. Three separate polar plots (Fig. 2, *A–C*) show that practically all three classes of excitability status (depicted by three different colors) are distributed throughout the 10-dimensional parameter space. Mean vectors in each of these cases are depicted by black dashed lines. For comparison, these three mean vectors and their corresponding standard deviations are plotted together in the polar plot of Fig. 2*D*. The Euclidean distance between the mean vector of excitable and the mean vectors of the other two solutions (nonexcitable or oscillatory) is ~ 0.15 , similar to the SD of distances within each of them (Fig. 2*D*, *Inset*). In other words, assuming complete independence of the parameters within the $\pm 25\%$ range of parametric variation, almost any randomly chosen vec-

tor of Hodgkin–Huxley parameters, regardless of its outcome (nonexcitable, excitable, or oscillatory), may be “pushed” to display any other excitability status by a minor manipulation of parameters.

Lower Dimension Hodgkin–Huxley Parameter Space. We focus on the conditions for transition between excitable and nonexcitable statuses. Several schematic momentary current–voltage relations of excitable membranes, during an action potential, are plotted in Fig. 3*A*. Grossly speaking, the lower curve depicts current–voltage relations sampled by voltage-clamp steps from deeply hyperpolarized holding potential. The upper curve depicts current–voltage relations sampled by voltage-clamp steps from a relatively depolarized holding potential. During an action potential, where membrane voltage is a dynamical free variable, current–voltage relations slowly shift between these two extremes due to an evolving voltage-dependent restoring force,

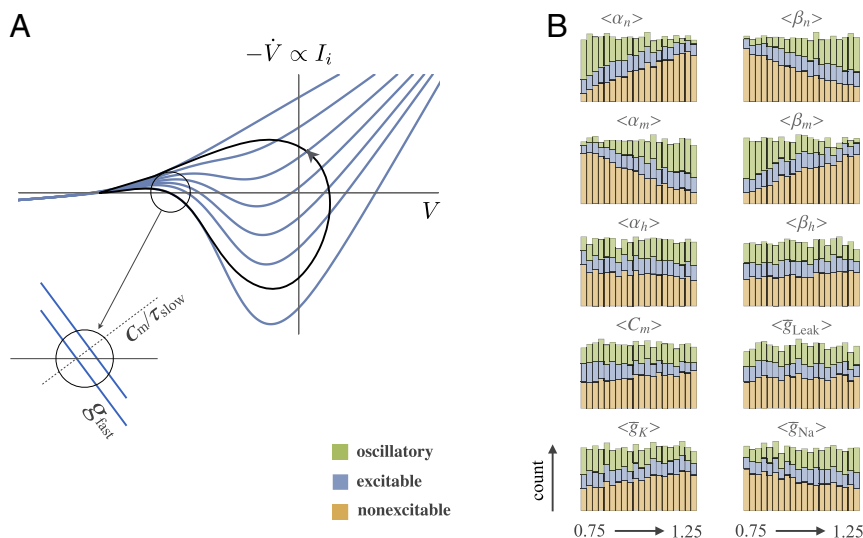


Fig. 3. Theory inspired dimensionality reduction. (A) Idealized momentary current–voltage relations at different ratios of available sodium and potassium conductances [see chapter 11 in Jack et al. (27)]. In different phases of the action potential, different momentary current–voltage relations determine the dynamics. The black continuous line depicts the resulting current–voltage trajectory during an action potential. (A, *Inset*) A magnified version of the area at threshold (indicated in the main image by a circle), about which the system is linearized. (B) Histograms of the three excitability statuses, constructed from the data of Fig. 2 (10,000 Hodgkin–Huxley realizations), for each of the scaling parameters. Note that all parameters are freely fluctuating, simultaneously, over $\pm 25\%$.

mediated by the opening of potassium channels and inactivation of sodium channels. The slow change in restoring force gives rise to a current-voltage closed trajectory depicted in Fig. 3A (black continuous line).

The general differential equation of the system is $C_m d^2 V / dt^2 + dI_i / dt = 0$. As pointed out by Jack et al. (27) (chapter 11), linearization about the threshold potential (Fig. 3 *A*, *Inset*) leads to an expression of dI_i / dt in terms of the momentary conductance ($-g_{\text{fast}}$) at threshold and the time constant (τ_{slow}) for evolving restoring force. The condition for instability near threshold, one of the solutions to this equation (expressed in conductance units), is $-g_{\text{fast}} > C_m / \tau_{\text{slow}}$. Note that these lumped entities ($-g_{\text{fast}}$, C_m , and τ_{slow}) may naturally be classified into the above groups of structural and kinetic parameters: C_m is obviously structural; likewise, the fast conductance ($-g_{\text{fast}}$) that depends on the relative contribution of maximal sodium conductance. In contrast, the time scale for introduction of restoring force (τ_{slow}) is a kinetic parameter because it depends on the actual transition rate functions governing the gating of sodium and potassium channels.

Inspired by the above and related mathematical reductions of excitability (24–26, 28), we turned to the data of Fig. 2 in search for these two dimensions, expressed in terms of Hodgkin–Huxley parameters. To this aim, we constructed count histograms of the three excitability statuses for each of the scaling parameters (Fig. 3*B*). These histograms show that the most critical determinants of excitability status are the rates of opening and closure of sodium and potassium channels ($\alpha_m(v)$, $\beta_m(v)$, $\alpha_n(v)$, and $\beta_n(v)$) and the maximal conductance of the membrane to the two ions ($\bar{g}_{\text{Na}}$, $\bar{g}_{\text{K}}$). (Similar results are obtained using the two “far apart” excitability statuses—nonexcitable and oscillatory—to extract principal components; see <https://www.wolframcloud.com/objects/marom/SM.nb>.) Distribution of excitability status is significantly less sensitive to transition rates involved in sodium conductance inactivation, as well as leak conductance. Membrane capacitance seems to have some effect. A possible interpretation of the histograms of Fig. 3*B* is that, at least as a first approximation, $-g_{\text{fast}}$ may be assumed to be proportional to a ratio of structural parameters $S = \langle \bar{g}_{\text{Na}} \rangle / (\langle \bar{g}_{\text{Na}} \rangle + \langle \bar{g}_{\text{K}} \rangle)$, whereas τ_{slow} may be assumed to be proportional to

the ratio of kinetic parameters $K = (\langle \alpha_n(v) \rangle + \langle \beta_m(v) \rangle) / (\langle \alpha_n(v) \rangle + \langle \beta_m(v) \rangle + \langle \alpha_m(v) \rangle + \langle \beta_n(v) \rangle)$.

The murky enmeshment of solutions shown in Fig. 2 (a sample of which is replotted in Fig. 4A, superposed) is significantly clarified when the data are arranged according to the values of S and K (Fig. 4B): The three different excitability regimes are nicely clustered in three clouds. The oscillatory and nonexcitable phases are well separated in the S - K plane (Fig. 4B, *Inset*), whereas the borders separating the excitable phase from these other two are “soft” rather than sharp. The nonexcitable cloud in the upper left corner is due to excessive sodium conductance that stabilizes the membrane at a depolarized potential.

Note that in Fig. 4B, four parameters (two sodium inactivation rates, capacitance, and leak conductance) are not taken into account, even though they are allowed to freely fluctuate. And, yet, when examined in the 2D S - K space, the Hodgkin-Huxley model reveals order that is literally impossible to detect in the more explicit, higher dimensional representation of Fig. 4A. The effectiveness of the dimensionality reduction is further supported in Fig. 5A, where the outcomes of multiple realizations of three different S - K pairs are shown. Each of these three predefined S - K pairs (0.60;0.45, 0.50;0.50, and 0.40;0.55) was realized 30 times by adjusting $\langle \alpha_m(v) \rangle$ and $\langle \bar{g}_K \rangle$ to the other four, randomly generated, Hodgkin-Huxley parameters ($\langle \alpha_n(v) \rangle$, $\langle \beta_m(v) \rangle$, $\langle \beta_n(v) \rangle$, and $\langle \bar{g}_{Na} \rangle$). Five of 30 are shown for each value; the rest are comparable. Clearly, the values of the lumped S and K dimensions are better predictions of the outcome than the individual Hodgkin-Huxley parameters.

Several points deserve attention in relation to the numerically calculated S - K plane of Fig. 4*B*. First, the borders between the three phases are fairly steep (note different ranges of S and K axes). The immediate implication of this steepness is that in its two extreme statuses (nonexcitable and oscillatory), the system is relatively immune to variations in maximal conductances. Stated differently, it is sufficient to use ionic channels that set the K dimension below ~ 0.45 to obtain a pacemaker that is insensitive to fluctuations in density of channel proteins; the system maintains its pace-making nature over factor 3 in the value of the structural (S) dimension. The same can be said about nonexcitable membranes: setting the K dimension above ~ 0.55

looked at channel densities (11, 19, 21). The present study suggests that when the problem is examined in a lower dimension, a simple control rule that relies on slow inactivation—a ubiquitous protein intrinsic process—can deal with fluctuations in both structural and kinetic parameters. This homeostatic mechanism is local, independent of protein synthesis, and operates over a wide range of time scales (milliseconds to many minutes).

We speculate that activity dependence of protein kinetics at relatively slow time scales, entailed by multiplicity of protein states, is a general “automatic” and local means for stabilization of cellular function.

Materials and Methods

All of the simulations and analyses were performed within the *Mathematica* (Wolfram Research, Inc.) environment. The *Mathematica* notebook

is available at <https://www.wolframcloud.com/objects/marom/SM.nb>. Data of Figs. 2–5 were generated using Hodgkin–Huxley equations, scaled to 6 °C, as they appear in the original manuscript (1952). The duration of each simulation epoch was 90 ms, including an initial 50-ms relaxation phase. Stimulus (7 μ A, 1 ms) was delivered 70 ms into the epoch. A sorting algorithm for excitability status (excitable, nonexcitable, or oscillatory) was constructed, which is based on the time and the number of spikes following the relaxation phase; spike detection threshold was set to 50 mV above resting potential. The sorting algorithm was validated by eye-inspection of multiple sets of 300 sorted epochs. To generate Fig. 6, we implemented an SVM algorithm with linear kernel within *Mathematica*, using an 80% training set.

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